Definition 12.10

A **statement formula** is an expression involving one or more statements connected by some logical connectives.

Definition 12.11: (Truth Table)

A table showing the relationship between truth values of simple statements and the truth values of compound statements formed by using these simple statements is called **truth table**.

Definition12.12

(i) Let p be a simple statement. Then the **negation** of p is a statement whose truth value is opposite to that of p. It is denoted by $\neg p$, read as **not** p. The truth value of $\neg p$ is T, if p is F, otherwise it is F.

(ii) Let p and q be any two simple statements. The **conjunction** of p and q is obtained by connecting p and q by the word **and**. It is denoted by $p \wedge q$, read as 'p conjunction q' or 'p hat q'. The truth value of $p \wedge q$ is T, whenever both p and q are T and it is F otherwise.

(iii) The **disjunction** of any two simple statements p and q is the compound statement obtained by connecting p and q by the word 'or'. It is denoted by $p \vee q$, read as 'p disjunction q' or 'p cup q'. The truth value of $p \vee q$ is F, whenever both p and q are F and it is T otherwise.

Logical Connectives and their Truth Tables

(1) Truth Table for NOT [$\neg$] (Negation)

Truth Table for $\neg p$

p	$\neg p$
T	F
F	T

Table 12.4

(2) Truth table for AND [$\wedge$] (Conjunction)

Truth Table for $p \wedge q$

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Table 12.5

(3) The truth tables for OR [$\vee$] (Disjunction)

Truth Table for $p \vee q$

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Table 12.6

12th_Maths_Vol 2_EM_CH 12_Discrete Mathematics.indd 239 16-12-2022 16:54:35